Other examples of 2-D array are:

int A[3][3];

float x[2][3];

char name[5][20];

12.3.2 Initialization of Two-Dimensional array

The array can be initialized in more than one way at the time of 2-D array declaration.

For example

```
int matrix[4][3]={
{10,20,30},// Initializes row 0
{40,50,60},// Initializes row 1
{70,80,90},// Initializes row 2
{100,110,120}// Initializes row 3
};
int matrix[4][3]={10,20,30,40,50,60,70,80,90,100,110,120};
```

Array's row size is optional but column size is compulsory.

For example

```
int matrix[][3]={
{10,20,30},// row 0
{40,50,60},// row 1
{70,80,90},// row 2
{100,110,120}// row 3
};
```

12.3.3 Accessing the two-dimensional array

Two-dimensional array uses two index values to access a particular element in it, where the first index specifies the row value and second index specifies the column value.

matrix[0][0]=10;// Assign 10 to the first element of the first row

matrix[0][1]=20;// Assign 20 to the second element of the first row

matrix[1][2]=60;// Assign 60 to the third element of the second row

matrix[3][0]=100;// Assign 100 to the first element of the fourth row

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